

A Companion Robot for Daily Care of Elders based on Homeostasis

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Abstract: This paper presents a hybrid system of applying homeostatic drives theory and stimuli from environment or human on a robot, aiming to solve the increasing need of elder caring in modern aging society. Through reinforcement learning with human feedback, the robot is enabled to make appropriate decisions by reviewing its own motivation, environment stimuli and human inputs, allowing the robot to finish vague tasks, *i.e.* to serve, help or have a conversation with human on different situations. In our experiment, the human satisfaction rate could reach 94% and the robot could as well maintain a stable condition to perform long-term service. If the robot is to be deployed in a nursing home, it can further serve and meet the requirements of elders, while simultaneously maintaining its own system, such as battery level, in regular daily lives.

Keywords: Elder care, robot service, Pepper, homeostasis, autonomy, dementia

1. INTRODUCTION

Recently, the elder population grows rapidly in most aging countries, and those with elder diseases like dementia need more medical cost. In Taiwan, the aging index has exceeded 100 in February, 2017 [1]; likewise, the elder population accounts for at least 27% of the Japanese population [2]. With the emerging problem and the heavy burden of the elder care, researchers have endeavored to bring domestic service robots to either caring centers or local households, which has long been a dream of human kind. For most modern service robots, they are usually embedded with a specific target which drives them to accomplish the specific service goals, and thus the sophisticated system implementation can be avoided, such as Roomba [3]. On the other hand, some approaches are using biologically inspired methods by which the robots try to imitate human's nervous system, and they become a new path for service robots' development [4].

In contrast, our objective here is to design a homeostasis based system with the ability to play nursing and companion roles, making the robot assist human to reduce the burden of elder caring. The homeostasis system is proposed from our previous work [5], which is a biologically inspired approach [6] that integrated the homeostasis drive theory [7] and the human intentions. The robot has inner motivations to trigger the aspiration to serve, and the ability to make decisions based on different situations of the surroundings and the user's intentions and actions. The model was trained by reinforcement learning, and used state mapping to be the planner at service. Since we aimed at the elder caring issue, we also redesign the performance of the robot to be more suitable for nursing and companion. The model was implemented on Pepper, a humanoid robot developed by SoftBank Group, Corp. [8, 9], and tested with our crews in the lab.

Based on our targets, we expected the robot can not only give assistance, but also give conversations, reminding, and provide entertainment. Moreover, the robot should also be able to deal with special

circumstances such as elders with dementia, giving a more thoughtful support as a whole. In addition, in order to be a long-term domestic elder caring robot, apart from providing appropriate service, it should be able to assess its own states and make necessary rest.

2. RELATED WORK

2.1 Homeostasis System

As mentioned, the model of previous work is a hybrid approach with homeostasis based system and human intentions. The former is the key to model internal needs, making the robot satisfy these needs in order to maintain a stable inner state, while the later takes the human into consideration, preventing the system from deciding actions by ignoring the human needs. Due to the collateral architecture of the model, Q-learning [10] is used to train the decisions based on internal (the service goal and life maintenance of the robot) and the external (human status and environment stimuli) parameters, finally forming a state-mapping model to act as the agent planner. The final goal of the model is to choose a proper action out of the predefined action set that will gain the maximum reward according to the current world state.

In our work, we substituted the human intention with human commands in order to give more straight-forward connection with the human to the robot. Besides, the actions are designed for the daily care of the elderly and those who suffered from dementia. Therefore, the purpose of an action is to satisfy the elder users or to meet the robot's internal needs when it comes to robot survival issue. Once the execution is done, all the parameters in both external and internal state will be updated.

2.2 Elder Caring and Robots

Due to the changes of the population structure, many researchers have turned the direction of ordinary service robots with specific goals to more comprehensive social and caring robots with vague targets. Moyle *et al.* [11] presented two vignettes to illustrate the benefits and challenges of testing social robots in the nursing home environment. They pointed out that dementia is one of

the major reasons why people enter nursing home (approximately 53%), and characteristics such as progressive impairment of an individual's functioning, decline in cognitive functioning, depressed mood, and altered communication are typical among these elders, causing their feeling socially isolated and lonely, and agitated behavioral problems such as sleep-wake and rest-activity pattern disturbances. The major challenges of setting social robots are poor Wi-Fi connection or the software system being out of action for a short time, and for the elders or the carers, the difficulty in learning how to set up and use the system. Hence, the designers must understand the perceptual, motor control and cognitive capabilities of users and, in this case, their carers who are the people often responsible for assisting with using the robot. Also, older adults in particular want a robot that looks friendly [12]. Huang et al. [13] introduced a virtual agent that can give conversational feedback and social signal, record speech and language information for assessing the cognitive status of the patient, and record memory vest which is equipped with portable devices (IC audio recorder, digital video recorder, and smartphone) to log the user's daily life. If the elder is lost, the agent will alert the user by the wireless locating system to keep safety [14]. Pollak [15] summarized current research trend of three assistive functions for elderly people with cognitive impairment:

1. Ensuring that the elder is safe and that he/she performs the necessary daily activities, and, if not, alerting a caregiver.
2. Helping the elder to compensate for his/her impairment by assisting in the performance of daily activities.
3. Assessing the elders' cognitive status.

As a result, by scanning ordinary elders' cognitive behaviors, we aspire to make the robot take care of their daily lives without influencing their habits; also, giving them assistance to maintain their basic life function. Most importantly, a self-functioning companion to give the elders warmth as a real human.

3. MODEL DESIGN

In our work, we developed a hybrid system with homeostasis drive theory [7] and stimuli from the environment and human to solve the daily care by well-implemented actions. By experimental evidence, the proposed system endows the robot interact with the elders actively and properly. The robot's needs and the user's commands are considered as situations in the system. The robot can learn its decisions to dedicate to elder caring and to self-sustaining at the same time through online reinforcement learning with human feedback. Furthermore, when robot detects some crucial events, it will react directly without taking its needs in to consideration. Base on this model, the extension and flexibility of the functions on such caring robot is promising.

The structure of our decision model mainly follows the structure in [5] as shown in Fig. 1. Q-learning is the main

algorithm to help the model trace everything in the scene. In advance, the model was trained with simulator to stabilize the transition factors, then the model was operated on real robot and kept learning while performing to adjust the real world. The process in each round can be divided into five parts: choosing motivations, detecting stimuli, checking crucial events, performing actions, updating model.

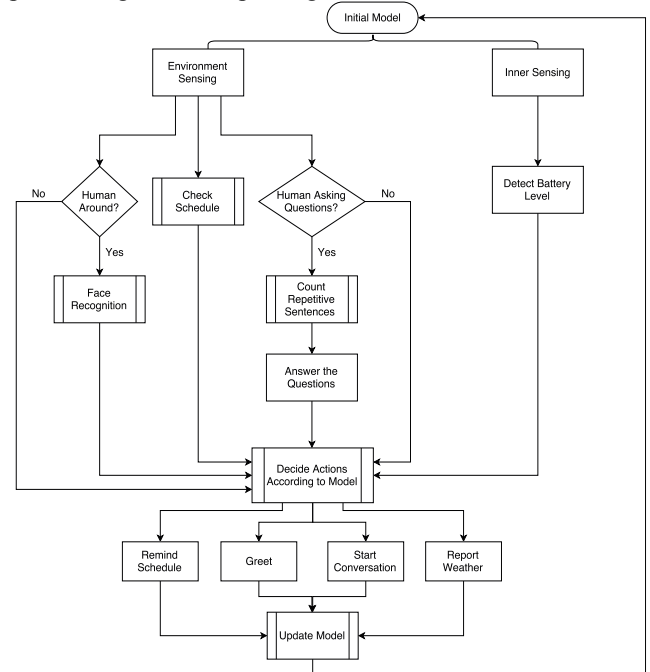


Fig.1 The structure of homeostasis system

4. METHODS

4.1 Robot Environment

Pepper (Fig. 2) is a humanoid robot, which is suitable for being an elder care. It has five types of motors, a tablet on the chest, and plenty of sensors, i.e. 2D cameras, 3D sensor, lasers, sonars, infra-red sensors, etc. With built-in system commands, Pepper is able to sense basic environment status such as human's presence or simple objects; moreover, it can present dialogs and gestures to interact with human. The programming language supports Python and C++, and its core library is NAOqi 2.4.3.28 [16].

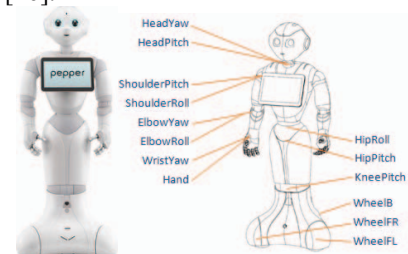


Fig.2 Pepper robot [16, 17]

4.2 Parameters Design

For an agent, the drives and motivations are the factors contributing to the internal status, while the stimuli play the role of affecting these internal states.

The drives (D) of an agent is a set of values

representing the internal needs, and a higher value of drive means a more severe need. Aiming at elder care, we chose the following drives to be implemented in our system:

1. Need of Achievement (NAch) [18]
2. Need of Socialization (NSoc) [19]
3. Need of Rest (NRes)

The motivations (M) of an agent is a set of goals for it to compensate for unsatisfied drives. Consequently, Pepper has three corresponding motivation:

1. NAch → Serving
2. NSoc → Social
3. NRes → Relaxative

The motivation of serving is to assist the elders to finish daily tasks, and the motivation of social is to actively present a conversation with the elder. At last, the motivation of relaxative is for self-maintenance, preventing from overheating or running out of battery. As an early testing model, we first design five actions which are assigned into three groups according to the motivations:

1. NAch: weather report, schedule reminder
2. NSoc: greetings, conversations
3. NRes: resting

Besides, upon starting environment sensing, a sequence of sub-processes would be activated including:

1. Face recognition
2. Check schedule
3. Count repetitive sentences

The dementia patients are more likely to forget the face and names of relatives. Hence, we use face recognition when someone comes into pepper's sight. Additionally, following a regular routine every day for dementia patients is very important. The schedule of each patient is stored and the alarm is set for important events. Moreover, repetition in speech or utterance is a scientifically proved classic symptom of dementia [20]. We use the dialogue system on pepper to track conversations between robot and human and calculate the repetitive sentences. If the patient asks some questions too many times, the robot will try to calm him/her down and try to answer the questions.

These actions are designed to be the specific service for dementia patients and have high privilege that need to be performed ahead of others. We aim to make their life more energetic and help the elders lifting up their moods.

4.3 Sensing

There are internal status of Pepper and external situations that will be collected by inner sensing and environment sensing; besides, we added the face recognition, schedule checking and counting repetitive questions as a feature in order to tackle above-mentioned issue. The followings are the stimuli we interest in:

1. Pepper's battery level
2. human's presence

3. recognized face name
4. incoming questions
5. human's repetition of utterance
6. schedule time

To observe internal status and external situations, we used the sensors built in Pepper's library. By accessing the memory, the robot's battery level showed from range 0 to 100 in point numbers; for sensing human, we turned on the sensors of camera, sound and touch detection. To detect nearby people possibly in sight, we set the built-in *basic awareness*, a status which Pepper will be attracted by surroundings and engage with the sources such as movement, objects or sounds. If a human was seen, a unique id and name would be stored in Pepper's memory. Meanwhile, Pepper would check if human's speech was input repeatedly and count the times of continuous repeating speech to detect dementia. At last, Pepper could also tell if there were schedules to be remembered.

4.4 Training and Testing

To save time and efforts, we use pseudo training to boost the whole process; that is, the beginning process was trained with simulator, giving the model random stimuli and feedbacks by its performance. The pseudo training took 5000 rounds, and within each iteration, Pepper would first check its inner dominant motivation and examine the external situations, then performed an action with the highest Q-value and get feedback from human. To simplify the process of giving feedback, we use the touch sensors on the robot as input. As Fig. 3 shown, the users can simply touch on robot's head, left hand and right hand to give positive, negative and dispensable feedback respectively.

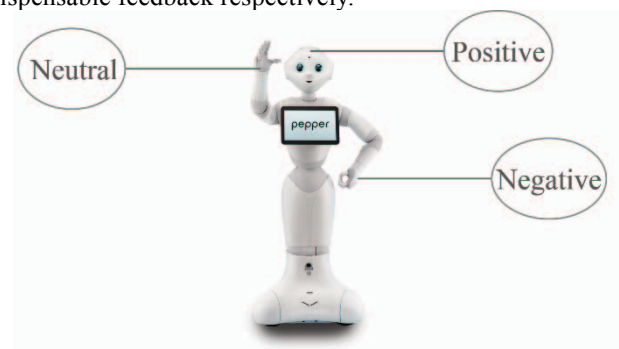


Fig. 3 The positions of touch sensors which represent different feedback.

At the end of each iteration, the reward gained would be used to modify the next round. After 5000 iterations of Q-learning, the state transformation in the model would not change frequently. We applied this state-mapping model on Pepper and tested it in real world. From interaction with human and gained feedbacks, the robot could serve the elders while learning, also having minor adjusts depending on different situations. Even though the human doesn't give the robot feedbacks, it can acquire feedbacks itself by calculating through the above-mentioned rules. The testing environment is set in the lab, with one member pretended to be the elder. The scene is about as large as an ordinary bedroom with no

obstacles around. To simulate the domestic situation, the testers would walk around and do daily tasks.

5. EXPERIMENT RESULTS

5.1 Simulation results

We used two indexes as the performance assessment:

1. Human Satisfaction Rate (HSR)
2. Secure Rate (SR)

The human satisfaction rate is the rate that the robot's action meets human's commands or the robot perform the correct action according to current situation. This parameter is user-related and defined as Eq. (1):

$$HSR = \frac{\# \text{ of positive feedbacks}}{\# \text{ of feedbacks}} \quad (1)$$

On the other hand, the secure rate is to describe the extent the robot feels secure. In this condition, any drive larger than 90 is considered insecure, resulting in the corresponding motivation has never been addressed, which is the last thing the robot would like to do. The SR is defined as Eq. (2):

$$SR = \frac{1}{N} \sum_{t=1}^N se(t),$$

$$se(t) = \begin{cases} 1, & \text{if } \forall d \in Drive: d(t) < 90 \\ 0, & \text{otherwise} \end{cases}$$

$$t: threshold \quad (2)$$

From simulation, Fig. 4 and Fig. 5 show the oscillating of the evaluation indexes. The changes of these parameters became stable after 5000 rounds. There was little prior knowledge provided to the virtual robot in the early stage of simulation, thus the fierce variation of states was found. As the simulation goes on, the robot gradually knew how to meet the human's needs thus the HSR kept rising. Meanwhile, the SR drops down and climbs up as the robot strikes a balance between satisfying human and itself. Though, there are some small oscillation follows, the SR still maintains in a tolerable range.

Experiment result shows a promising performance after learning, reaching about 94% human satisfaction rate and about 98% robot secure rate, which means the model could effectively satisfy the human needs according to different situation with its own status stabilized. Since the model learned during operation, the detailed oscillation of each parameter would have minor differences in each experiment.

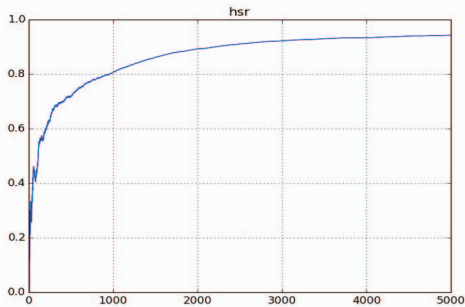


Fig.4 Changes of HSR

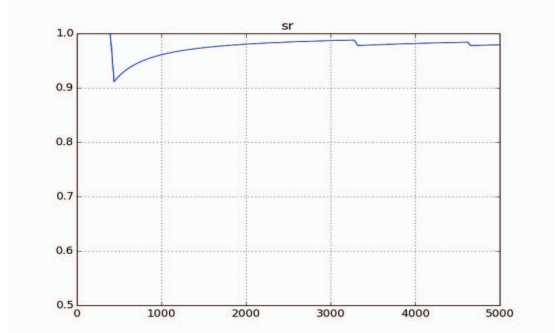


Fig.5 Changes of SR

5.2 Testing results

After the simulation was done, we set up an experiment with one tester in an open space with the daily care robot to test the performance, shown as Fig. 6. The experiment is aimed to test whether the robot is able to deal with the following task:

1. Recognize the face and say the name of the incoming person to greet him/her.
2. Answer the following questions:
 - a. Where is my child?
 - b. When should I take the pill?
 - c. Where is the key?

The answer of a. is provided by checking the child's schedule, for b. is checking the patient's schedule, and for c. is searching nearby area by its vision.

3. Track the repetitive utterance and calm down the patient if needed.
4. Provide the actions (greetings, conversations, weather report, schedule reminder, resting) from the model properly.

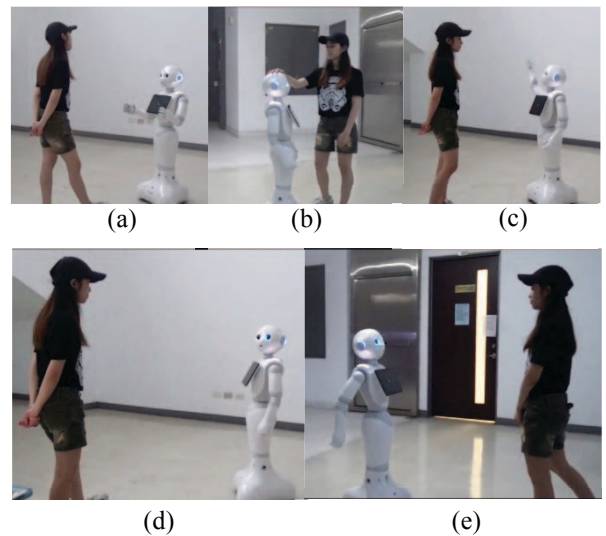


Fig.6 Pepper and human interaction in the experiment.

(a) Pepper answering question. (b) human giving feedback. (c) Pepper presenting greeting. (d) human tracking and face recognition during service. (e) Pepper forwarding to human after scanning.

Evaluating the experiment by neutrality, there are some pros in this model:

1. The accuracy of face recognition is about 80%, evaluated by built-in function of Pepper.
2. Once Pepper recognizes the dialogue, it can reply the right answer to the elders. Also, Pepper can tirelessly answer the repetitive questions and calm the patient down.
3. For making the elders' life more energetic, the conversation Pepper provides can please the elders and reduce the loneliness.

Whereas there exists some cons:

1. The dialogue system of Pepper has limited ability, causing the conversation topic restricted. We need to build up the database to enlarge the topic.
2. The actions are not diverse enough. Hence, we can only address the simple tasks now.

6. CONCLUSION

Recently, Pepper is gaining popularity for either commerce advertising or service guidance. Expecting its appearance will flourish before long, we aspire that this model is a solution to tackle the elder caring problem, which is an inevitable and costly social issue, helping to alleviate the stress of modern countries and simultaneously letting the elders observe thoughtful care without feeling abandoned and activating the brain activities of patients with dementia. Currently, our homeostasis model is capable of dealing with single elder or patient, with stable performance and efficiency. However, the model provides service mainly on dialogue system, which is a bit insufficient to put it in practice. The next step of our work is to add more functions and test the model in caring centers to contact with multiple real elders.

Now we are developing more complicated action sets to the homeostasis system, and expecting to add them to our system before long. The first one is fruit recommendation system. In order to improve the nutrients balance or prevent the elders from possibly damaging their vulnerable teeth, the robot will choose suitable fruit depending on the users' situation and remind them to absorb necessary sources. Second, the scenario conversation system enables the robot to scan the user's surroundings and movements, and start a conversation with the content of objects nearby the human or the actions that the human is doing. This better the interactions with the robot and the human, giving a more humanoid feeling. With the above-mentioned features, we believed that having a convenient domestic service robot will no longer be a dream.

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